

- electromagnetic receptors and, 1111f
as fixed action pattern, 1140, 1141f
genetic variation in patterns of, 1156, 1157f
movement corridors and, 1271f, 1272
- Milk, mammalian, 741–742, 743f, 1005f–1008f, 1016
- Milk duct, 392f
- Milkweed, 645f, 653
- Miller, Stanley, 57f, 58, 526f
- Millipedes, 708, 709f
- Mimic octopus, 1219f
- Mimicry
endorphin, 1081
molecular, 40f, 78
in prey and predator adaptations, 1218f–1219f, 1237
as prey defensive adaptation, 868f
- Mimivirus, 408
- Mimosa pudica, 802, 862f
- Mimulus species, 522f
- Mineralized dental elements, 724f, 725
- Mineralized organic matter, 529f
- Mineralocorticoids, 1004f, 1012f, **1014**
- Minerals, 899f, **900**, 901t
deficiencies of, in plants, 810, 811t, 812
mineralocorticoids and metabolism of, 1014
mycorrhizae and plant deficiencies of, 818
root architecture and acquisition of, 787
roots interaction with, 10f, 11
transpiration of, from roots to shoots via xylem, 792, 793f–795f, 796
vascular plant transport of, 787, 788f–791f
- Mines, restoration of, 1253f
- Miniaturization, gametophyte, 637f, 638
- Minimal medium, 336f–338f
- Minimum viable population (MVP), **1268**
- Minnnows, 1143f, 1276
- Mirounga angustirostris*, 999f, 1267
- Miscarriages, 306, 1039
- Misfolding, protein, 83
- Mismatch repairs, **327**
- Missense mutations, **357**, 358f
- Mississippi River, 1275f
- Mistletoe, 819f
- Mitchell, Peter, 177
- Mites, 707, 708f
- Mitochondria, **109**
animal cell, 100f
animal hibernation and, 178
in apoptosis, 230f, 231
ATP synthase and, 186
chemical energy conversion by, 109–110, 111f
chemiosmosis in, 175, 176f, 177, 199f–200f, 201
dinitrophenol and, 186
electron transport chains in, 168–169
endosymbiotic origin of, 534, 535f
enzymes in, 161f
evolutionary origins of, 109, 110f
fungal cell, 100f
inheritance of genes of, 311
origin of, in endosymbiosis, 595, 596f–597f
plant cell, 101f, 209f
protist, 594
pyruvate in, 171f
using cell fractionation to study, 97
- Mitochondria DNA (mtDNA)
evolutionary rate of, 565
species identity in, 557f
- Mitochondrial matrix, **110**
- Mitochondrial myopathy, 311
- Mitosis, **236**. *See also* Cell cycle; Cell division
in animal cells, 238f–239f
in chromatin packing, 332f
in daughter cells, 234
evolution of, 243, 244f
in human life cycle, 257f, 258
information in, 252
meiosis vs., 262, 263f, 264
nuclear envelope during, 252
origin of term for, 237
in plant cells, 241, 242f–243f
in spermatogenesis, 1042
in varieties of sexual life cycles, 258f, 259
- Mitosomes, 600f
- Mitotic (M) phase, **237f**
- Mitotic spindles, **240f**–241f
- Mixotrophs, **594**
- Mobile genetic elements, evolution of viruses and, 408
- Mobile River basin, 1264
- Model organisms, **22**, **1044**. *See also* *Drosophila melanogaster*; *Escherichia coli* bacteria
Arabidopsis thaliana, 22, 774, 776f, 783
bread mold. *See* *Neurospora crassa*
Caenorhabditis elegans, 22, 705
in developmental biology, 1044
for DNA research, 316f, 317
mouse (*Mus musculus*), 22. *See also* Mouse
Neurospora crassa, 664f, 665t
scientific cooperation and, 22–23
for T. Morgan's experiments, 295f
- Models
atomic, 30f
community disturbance, 1228, 1229f
of covalent bonds, 36, 37f
electron orbital, 35f
exponential population growth, 1196, 1197f, 1207f, 1208
island equilibrium, 1232, 1233f
logistic population growth, 1197, 1198f–1199f, 1198t, 1200
molecular-shape, 39f
optimal foraging, 1149f
process of science, 19f
testing hypotheses with quantitative in Scientific Skills Exercise, 1150
- Modified leaves, 762f
- Modified roots, 760f
- Modified stems, 761f
- Mojica, Francisco, 253
- Molarity, **50**
- Molar mass, 50
- Molar ratios in Scientific Skills Exercise, 58
- Mold model organism. *See* *Neurospora crassa*
- Molds, **658f**–659f, 662f, 663
- Molecular basis of inheritance
chromatin packing of DNA and proteins in eukaryotic chromosomes, 330f–332f
discovery of double helix structure of DNA in, 314f, 317, 318f–320f
DNA replication and repair in. *See* DNA replication
evidence for DNA as genetic material in, 315f–317f, 318
evolution of gene concept from, 361
- Molecular biology
Arabidopsis thaliana as model organism for, 774, 776f
determination of microbial diversity using, 1223f
importance of viruses to, 400
measures of evolution in, 87, 89
mutants in, 845
of plant development, 775f–780f
- Molecular clocks, **566**, 567f–568f, 659f, 660
- Molecular formulas, 36, 37f, 59f
- Molecular genetics, in ecological forensics, 1265f
- Molecular genetic variation, 487, 488f
- Molecular homologies, 479–480, 558, 559f
- Molecular identification tags, 107
- Molecular-level herbivore defenses, plant, 868f
- Molecular mass, **50**
- Molecular recognition, immune system, 953, 957f
- Molecular systematics, 563f, 583, 682, 683f, 684.
See also Cladistics; Systematics; Taxonomy
- Molecular tags, 332f
- Molecules, **5f**, **36**. *See also* Compounds
biological. *See* Biological molecules
chemical bonds and formation of. *See* Chemical bonds
as level of biological organization, 5f
organic. *See* Organic compounds
origin of self-replicating, 526f–528f
regulation of interactions of, 10f
shape and function of, 39f–40f, 59f
structure of DNA and RNA, 7f–8f
- Moles, 1107f–1108f
- Moles (animal), 558f, 559
- Moles (mol), **50**, 58
- Molluscs, 688f
bivalves, 699, 701f–702f
body plan of, 699f
cephalopods, 699, 701f, 702
chitons, 699f
in Ediacaran period, 535f, 536
eye complexity in, 547f–548f
gastropods, 542, 699, 700f–701f, 702
nervous systems of, 1086f
- protecting freshwater and terrestrial, from extinction, 702f
- Moloch horridus*, 737f
- Molothrus ater*, 1271, 1287
- Molting (ecdysis), 683–684, **705**, 707, 894f
- Monarch butterflies (*Danaus plexippus*), 839, 1146f
- Monera kingdom, 568
- Monilophytes, **622**, 622t
- Monkey flowers, 522f, 834
- Monkeys, 56f, 87, 89, 746f–747f, 1122f. *See also* Chimpanzees
- Monocilia, 1064
- Monoclonal antibodies, **969**
- Monocots, **649f**–650f, 761, 769f–770f, 829f–830f
- Monocytes, 934f–935f
- Monod, Jacques, 366
- Monogamous mating, 1024, **1149**–1150, 1151f
- Monoglycerides, 910f
- Monohybrid crosses, **274**
- Monohybrids, **274**
- Monomers, **67f**
- Monophyletic groups, **560f**
- Monosaccharides, **68f**–69f
- Monosodium glutamate, 1123
- Monosomic cells, **307**
- Monosomy X, 309
- Monotremes, 543f, **742**, 743f, 745f
- Monozygotic twins, 1036, 1061
- Monterey County, California, 614f
- Montmorillonite clay, 527, 528f
- Montreal Protocol, 1284
- Moon jelly, 922f
- Moose, 1205f
- Moray eel (*Gymnothorax dovii*), 729f, 1214f
- Morels, **663f**, 670
- Morgan, Thomas Hunt, 295f–296f, 301f–303f, 315
- Mormon tea, 642f
- Morning-after birth control pills, 1038f
- Morning sickness, 65, 1036
- Morphine, 40f, 78
- Morphogenesis, 381–**382**, **1049**. *See also* Embryonic development; Pattern formation
apoptosis in, 1056–1057
cytoskeletons in, 1056f–1057f
developmental adaptations of amniotes in, 1053f, 1054
gastrulation in, 1044, 1049, 1050f–1053f
organogenesis in, 1044, 1049, 1054f–1055f
plant development and, 776–777, 778f
- Morphogen gradients, 386
- Morphogens, **386f**–387f
- Morphological homologies, 558
- Morphological isolation, 508f
- Morphological species concept, **510**
- Morphology
animal phylogeny and, 682, 683f, 684
fungal, 655f–656f
macroevolution of, 544f–546f, 547
species concepts and, 507
- Mortality rates. *See* Deaths
- Morton, Michael, 24f
- Mosaicism, 300f
- Mosquitoes, 409, 412f, 485, 496, 503f, 518, 606f, 712f, 713
- Mosquitofish (*Gambusia hubbsi*), 511f, 512
- Mosquito larva, 342f
- Mosses, 621f, **623**, 624f–627f, 628, 635, 637f. *See also* Bryophytes
- Mossy leaf-tailed gecko, 26
- Moths, 43, 476f, 551, 712f, 717, 824f, 1001, 1006f, 1007, 1110, 1111f
- Motile cilia, 1064
- Motility, prokaryotic, 576f, 577
- Motor, flagellum, 576f, 577
- Motor areas, cerebral cortex, 1096
- Motor cortex, 1097f, 1098
- Motor neurons, 1068, 1069f, **1088f**, 1128, 1129f–1130f
- Motor output stage, 1068f
- Motor proteins, 76f, **113f**, 115, 116f, 123f, 152f, 241f, 252
- Motor systems, **1088**
cardiac and smooth muscle in, 1131–1132
muscle function in, 1125, 1126f–1131f, 1132
sensory systems and, 1107f–1108f. *See also* Sensory systems
skeletal muscle contraction in, 1127f–1131f
skeletal systems and locomotion in, 1132f–1135f, 1136